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The role of emergent information technologies and systems in enabling supply chain agility

A. White^a, E.M. Daniel^{b,*}, M. Mohdzain^c

^a*Cranfield Centre for Logistics and Supply Chain Management, Cranfield School of Management, MK43 0AL, UK*

^b*Centre for Innovation, Knowledge and Enterprise, Open University Business School, MK7 6AA, UK*

^c*Information Systems Research Centre, Cranfield School of Management, MK43 0AL, UK*

Abstract

In many markets it is becoming impossible to remove or ignore sources of turbulence and volatility. Hence, supply chain managers must accept uncertainty, but still need to develop a strategy that enables them to match supply and demand at an acceptable cost. The ability to achieve this has been termed supply chain agility. Information and, importantly, agile information systems have been recognised as being a critical factor in achieving agility in the supply chain.

To date studies have suggested that high levels of coordination between organisations, necessary for improved supply chain performance, demand high levels of integration between partners' information systems. However, these high levels of integration reduce the ability to frequently and rapidly make changes to trading relationships, something that is a prerequisite of the agility paradigm. This paper seeks to explore how a number of emergent information systems offer the possibility of both deep integration and increased flexibility. This paper is exploratory in nature. A single case study of an organisation seeking to improve supply chain agility is reported. Existing literature and the insights provided by the case study are combined to identify a number of themes for further study.

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Keywords: Supply chain agility; Information systems; Web services; e-Hubs

*Corresponding author. Tel.: +44 1908 654572.

E-mail addresses: Andrew.White@cranfield.ac.uk (A. White), E.M.Daniel@open.ac.uk (E.M. Daniel), mohdzain@cranfield.ac.uk (M. Mohdzain).

1. Introduction

Organisations in a diverse range of markets are facing the challenges of new product innovations, decreasing product lifecycles, product proliferation and customers who are becoming ever more demanding. This has resulted in markets that can be characterised as increasingly turbulent and volatile (Brown & Eisenhardt, 1998) and has caused many organisations to seek to improve their supply chain through activity termed supply chain management (Christopher, 1998; Cooper & Ellram, 1993; Mentzer et al., 2001). The operation of an appropriate and effective supply chain is now a vital part of competitive performance in all types of markets (Subramani, 2004).

To date, most supply chain management activities have sought to remove or reduce the uncertainty within a supply chain as far as possible, in order to facilitate a more predictable response to changes in downstream demand (Mason-Jones, Naylor, & Towill, 2000). However, in many underlying markets it is becoming impossible to remove or ignore sources of turbulence and volatility. Hence, managers of supply chains are faced with the situation where they have to accept uncertainty, but where they still need to develop a strategy that enables them to match supply and demand. The ability to manage such turbulence, in a manner that responds to customer requirements but at an acceptable cost, has been termed supply chain agility (Christopher, 2000). Recognising that market turbulence is unlikely to abate, developing an agile supply chain is now a major focus of many leading organisations (Fisher, 1997).

Whilst a number of diverse prerequisites have been identified for achieving supply chain agility including; management commitment, organisational and supply chain design and information sharing both within and between organisations (Bal, Wilding, & Gundry, 1999; Crocitto & Youseff, 2003), the use of effective information systems has been identified as a key enabler of supply chain agility (Garcia-Dastugue & Lambert, 2003). As described by Breu, Hemingway, Strathern, and Bridger (2001), “information systems are seen to assume a fundamental role in developing agility, as the notions of speed and flexibility would be inconceivable without them”. Over the last decade many companies have invested in large integrated information systems, such as enterprise resource planning (ERP) and customer relationship management (CRM) systems in order to improve, inter alia, their supply chain operations (Akkermans, Bogerd, Yucesan, & van Wassenhove, 2003; Davenport & Brooks, 2004). However, whilst the deployment of such systems may have improved the efficiency and effectiveness within their own organisation, the diverse nature of these systems has often reduced the ability to form electronic linkages with trading partners (Crowe, 1992; Davenport, 2000). In contrast, an important part of the agility paradigm is the ability to form deeply integrated links with a wide range of trading partners and be able to quickly dissolve these and reform such deep linkages with new partners as required by changing market conditions.

To counter the recent focus on internal information systems, it is forecast that over the coming decade emphasis will shift from the individual firm to improving collaboration between organisations (Hagel & Seely Brown, 2001). However, this improved collaboration must allow for the increased agility in the supply chain that organisations are seeking. A number of recent developments in information systems and technology (IS/IT), if used effectively, appear to promise the ability to significantly improve the collaboration between trading partners in ways that can increase agility. This paper seeks to explore the use of such emergent information systems

in enabling supply chain agility. Due to the emergent nature of these developments, this paper is exploratory in nature and its intention is to encourage further study of this domain. A single case study of an organisation that is seeking to improve the agility of its supply chain is undertaken. Existing literature from the inter-organisational (IOS) domain and the insights provided by the case study are combined to identify a set of themes that can frame future studies.

This study will be of interest to both supply chain and information systems managers who are seeking means to improve the agility of their supply chains. The synthesis of the most recent thinking in the supply chain management domain, with emergent developments in information systems, will also make this paper of interest to academics from both of these fields, together with academics from other fields interested in the improvement of inter-organisational relationships (IOR).

The paper commences with a discussion of supply chain agility. The role of information systems in the supply chain is then discussed. The case study is then presented and the insights from this, together with existing literature from the IOS domain, are used to identify a number of themes that could provide the basis of future studies.

2. Agility in the supply chain

The concept of agility within organisations originated within the function of manufacturing and with the concept of flexible manufacturing systems (Bennett, Forrester, & Hassard, 1992; Parthasarthy & Prakash, 1992). Agile manufacturing can be viewed as the antithesis of the characteristics that came to define manufacturing during most of the 20th Century: rationalisation, standardisation and the elimination of uncertainty (Womack, Jones, & Roos, 1990). In many industries today, the prerequisites of successful manufacturing are organisations, processes and products that can sense and change or be changed in response to customers' varying demands (McCarthy & Tsinopoulos, 2003).

As organisations continue to seek competitive advantage, the focus for the source of this has expanded from the realm of the single organisation to the various supply chains that they are actors within. The concept of agility has also moved from within the organisation to the field of supply chain management. The agile supply chain is an approach for managing supply chains in volatile markets (Naylor, Mohamed, & Berry, 1999). It can be defined as “a business-wide capability that embraces organisational structures, information systems, logistics processes and, in particular, mindsets” (Christopher & Towill, 2002a).

Agile supply chains have been characterised as: utilising virtual teams (Bal et al., 1999); having time compressed business processes (Mason-Jones & Towill, 1999); communicating real-time market data via information systems to all parties in the supply chain (Christopher & Towill, 2002b); making use of contract manufacturers (Mason, Cole, Ulrey, & Yan, 2002); being responsive to changes in throughput, destinations and volumes (Huang, Uppal, & Shi, 2002; Prater, Biehl, & Smith, 2001); and the use of de-coupling and postponement points (Mason-Jones et al., 2000; van Hoek, 2000).

As shown in Fig. 1, information and, importantly, agile information systems are recognised as being a critical factor in achieving agility in the supply chain (Christopher, 2000) in a similar way to the contribution flexible manufacturing systems made in the past to agile manufacturing.

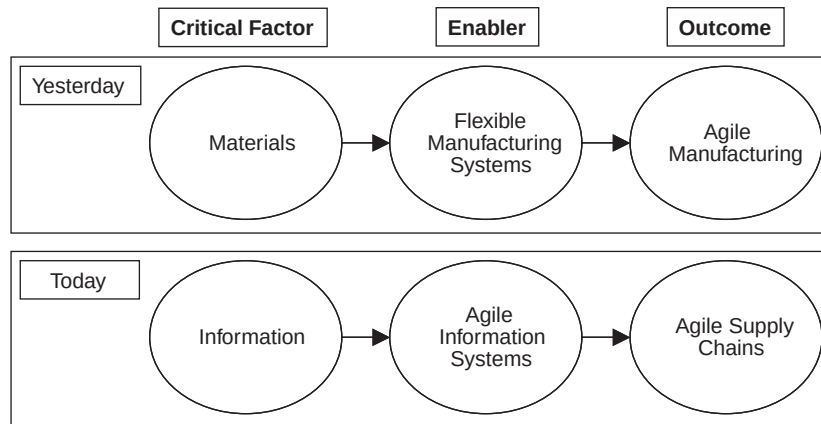


Fig. 1. Evolution of the agility paradigm (based on Christopher, 2000).

3. IS/IT in the supply chain

The extant literature addressing the development and use of IOS is considerable (for example; Angeles, Corritore, & Basu, 2001; Bytheway & Dhillon, 1996; Holland, 1995; Webster, 1995; Williams, 1997). It is therefore not possible to give a full review of this.

Hong (2002) defines IOS as “the information and communications technology that transcends organisational boundaries”, continuing on to say that IOS “function to blur the boundaries of today’s organisations as they enable information to flow from one organisation to another”. As information technologies have permeated the business arena over the past four decades, many different IOS have developed in diverse industries such as airlines, healthcare and the automotive industry among others (Williams, 1997). Typical examples of IOS would include systems such as electronic data interchange (EDI), electronic funds transfer (EFT) and systems, such as supply chain management systems, based upon extensible mark up language (XML).

Extant studies suggest that, the greater the degree of coupling or integration between the information systems of trading parties, the greater the degree of coordination and collaboration that can be achieved (Goodhue, Wybo, & Kirsch, 1992; Holland, 1995). This suggests that firms wishing to improve the performance of their supply chain operations should invest their time and resources in establishing closely coupled links between themselves and their trading partners. Indeed, McLaren, Head, and Yuan (2002) describe how shared systems, that is systems that are jointly owned between the trading parties, or which one firm has made available to others, represents the highest degree of integration between individual firms and also the greatest expected benefits.

However, as observed by Clemons, Reddi, and Rowe (1993), the formation of such close information system linkages with other organisations involves relationship-specific investments. Although the costs of IT are reducing significantly and there is an increased adoption of open standards, both of which would suggest a reduction in the degree to which investments are relationship specific, there are still significant costs associated with the establishment of human relationships and with the redesigning of business processes, which will be specific to individual

relationships. This suggests that there are transactional economies of scale which argue for fewer trading partners, but with whom the organisation maintains longer relationships. This forms the basis of their 'move to the middle' or 'suppliers-as-partners' hypothesis. The requirement for relationship-specific investments may be expected to increase with the degree of integration established and will therefore be greatest in tightly coupled linkages (Allen & Boynton, 1991; McLaren, et al., 2002). Firms are therefore unlikely to be willing to change these highly integrated relationships, suggesting a lack of flexibility. This suggests that, IOS integration and IOS flexibility are linked and are inversely related, that is an increase in integration reduces flexibility and vice versa (De Leeuw & Volberda, 1996; Duncan, 1995).

The notion of agility in the supply chain suggests the need to be able to develop close linkages with trading partners, ideally supporting a number of key processes that will improve the responsiveness of the organisation including procurement, forecasting, supply chain management and new product development (Christopher, 2000). However, for reasons such as the phasing out of older products and the introduction of new products, the expansion into new geographic markets or a growth in demand, it may be necessary for extant linkages with existing trading partners to be rapidly dismantled and new, equally close relationships formed with different partners.

A number of new information systems and technologies have emerged over recent years that may be able to change the relationship between IOS integration and flexibility. A key technology is that of web services. Web services encompass a set of interrelated technologies that allow information systems to be assembled from components that can easily be integrated with each other, regardless of the programming language that is used to create those components or the platform upon which they run (Fensel & Bussler, 2002; Hagel, 2002). They can therefore allow organisations to easily form integrated linkages with suppliers who may be operating different internal systems (Vidgen, Francis, Powell, & Woerndl, 2004; Wu, 2004). Whilst web services offer many benefits in this domain, their use with other emergent technologies and systems appears to offer the greatest opportunity for improved agility. Other technologies that may be expected to contribute include: electronic trading hubs, to facilitate multi-party information sharing and trading (Sawhney, 2000); business process management systems, to orchestrate inter-organisational processes (Leymann, Roller, & Schmidt, 2002) and automatic data capture, to allow the reconciliation of the physical location and tracking of goods with information flows (Fisher, 1997).

4. Methodology

The objective of the current study was to explore the potential of new information technologies and systems in enabling supply chain agility. This early stage exploration, together with existing literature from the IOS domain, would be used to frame a set of research themes that can be used to guide future studies in this domain.

The current study seeks to explore a new domain and from this exploration commence the process of theory generation. This suggests the use of inductive, qualitative methods rather than deductive, quantitative research methods more appropriate for theory testing (Hussey & Hussey, 1997; Locke, 2001). Such inductive methods are frequently operationalised as case studies

Table 1
Case study interviews undertaken

Organisation	IBM integrated supply chain division	E2Open
Industry sector	Computer hardware and software	Integration hub for electronics industry
Supply chain staff interviewed	5	3
Information systems staff interviewed	3	1
Location of interviews	USA and UK	USA
Total number of interviews	8	4

(Eisenhardt, 1989; Yin, 1989), and this approach was therefore adopted. The value of the case study approach lies in its ability to consider theory in the context of the rich picture of the organisation studied, including its unique idiosyncrasies (Hoskisson, Hitt, Wan, & Yiu, 1999).

The case studied was that of the supply chain of IBM. IBM was selected for study since it represents a highly successful company operating in a volatile and turbulent market, that of computer hardware and software. Recognising its importance to future success, IBM has recently made significant efforts to improve the operation of its supply chain activities. This includes making use of two of the emerging information systems and technologies discussed above, that of an independent e-hub for inter-organisational integration and via this hub, the use of web services.

In addition to interviewing key staff within the supply chain and information systems operations of IBM, interviews were conducted with key staff in the e-hub, E2Open. E2Open was studied since IBM is making increasing use of this e-hub. However, E2Open is interesting in its own right. It is very successful at user recruitment, with over 1500 companies worldwide using it for inter-organisational process integration. Also, an analysis of leading electronic hubs, providing both procurement and other services undertaken across all industry sectors (Metcalf, Meringer, & Roitman, 2002), described E2Open as “leading the pack”.

The interviews undertaken are shown in Table 1. In total twelve interviews were carried out, both in the UK and in the USA. Each interview lasted between one and two hours and was tape-recorded. A summary of the interview findings was written up as a case history, which was passed back to the interviewees in order to check for accuracy and completeness, thus providing a source of internal validity for the findings. The interview findings were supplemented by supporting documentation both from within IBM and E2Open and from external commentators, contributing to external validity.

5. The case of IBM integrated supply chain division

IBM is a global leader in the development and manufacture of information technologies, which include computer systems, software, networking systems, storage devices and microelectronics. IBM had annual revenues in 2003 of \$89bn, of which \$43bn was from IT services, \$28bn from hardware, \$14.3bn from software and \$3bn rental and financing. It has 319,000 employees in 170 countries and its head offices are located in New York State.

5.1. Becoming an ‘on-demand’ business

In their Annual Report for 2003, IBM set out their intention of increasing the agility of their entire organisation. They described how they wished to be able to respond to the volatility and turbulence in their markets by becoming what they described as an ‘on-demand’ business. Such a business they believed had processes that are: responsive to any client demand, market opportunity or external threat; integrated end-to-end across the company and integrated across their partners, suppliers and clients.

IBM operates within a complex supply network that handles over 78,000 products, with over 3 million configurations. A key part of meeting their vision of becoming a highly responsive organisation was to improve the agility of this supply network. In January 2002 an Integrated Supply Chain Division (ISCD) was established to build “the most advanced and cost-effective on-demand supply chain in the industry”. The division works across multiple business units within IBM as shown in Fig. 2. The ultimate aim is for a customer to be able to place a single order for all of their goods from IBM, and suppliers to have a single point of reference within the organisation.

Whilst the ISCD employs a number of means of communicating with suppliers and customers, including phone, fax, extranet and EDI, a major initiative for them is the use of the electronic hub E2Open, as depicted in Fig. 3.

E2Open was established in 2000 by a consortium of leading organisations in the electronics industry. Originally established as an electronic trading marketplace (Baljko, 2003), like many such organisations, E2Open found that establishing a viable trading community proved challenging (Miller, 2001; Wise and Morrison, 2000) and like other marketplaces they sought to evolve their offerings (Laseter and Capers, 2002). E2Open has since positioned itself as a third party hub between organisations in the electronics industry that wish to achieve integration between their information systems.

5.2. Operation of E2Open

E2Open is based upon a ‘software-as-a-service’ model. It operates a platform and set of supply chain applications, which organisations in the electronics industry can use, as they need them, to connect with their customers and suppliers who are also using E2Open. The services can be

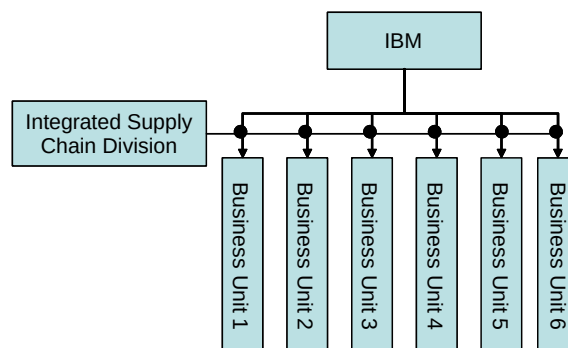


Fig. 2. Integrated supply chain division.

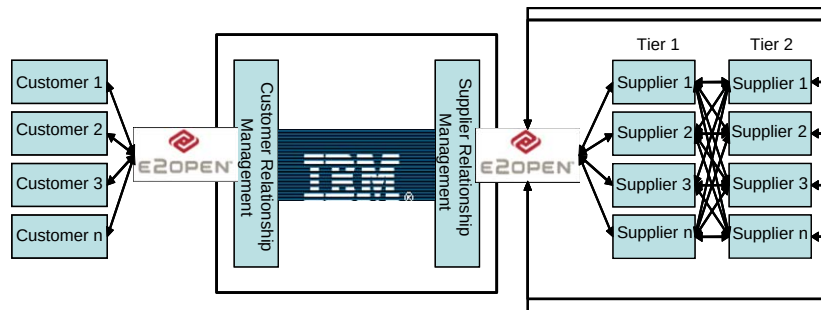


Fig. 3. IBM's use of E2Open.

accessed by a browser or can be integrated into an organisation's existing back-end systems. E2Open offers systems that support inventory and order management, including request for quotations (RfQs) and materials content information, and the tracking of orders, shipments and deliveries.

The services and applications offered by E2Open, and consequently those used by IBM are based upon web services. Use of web services allows IBM to publish to its trading partners via the UDDI directory hosted by E2Open, details of and access to its on-line processes, such as purchase orders and RfQs. This directory forms the intelligence behind the any-to-any integration enabled by E2Open and is described as the "switch board for the network". IBM's suppliers and customers also making use of E2Open can easily make use of these services to exchange information and trade with IBM.

IBM are using E2Open's services to improve communications with both their customers and suppliers, that is for both the management of supply and demand as shown in Fig. 3. An important attraction of the many-to-many model of E2Open is the ability for multiple organisations in the supply chain to connect to each other, allowing improved performance of the entire chain since "information from any tier in the supply chain is available to any other organization in the chain".

In order to support many-to-many trading, E2Open provides various support or enabling services. These include security, registration and authorisation procedures, legal documentation and high-level business process information such as standard business process maps, for example, RosettaNet Partner Interface Process (PIP[®]) maps. They also provide translation services between existing standards:

We also see ourselves being able to add value to our customers operations through providing secure transport of information as well as any-to-any translation. So you might send a document in the EDI Odette standard and your partner might require it in X12, so we will facilitate the translation back and forth.

5.3. *Providing the enablers of agility*

The use of E2Open by IBM and its web service-based applications enables a number of the vital components of the agility paradigm. An important aspect is that rather than encourage the

development of disparate systems within different organisations within a supply network; E2Open's shared services encourage standardisation across the industry. This in turn allows IBM, or other industry participants, to easily form electronic linkages with their trading partners and to reconfigure these when market conditions require.

Because this is a shared product, which is being driven by the industry, rather than just IBM, we will be moving to a more industry standard environment. We are looking at the ability of the software to manage the supply chain, as opposed to our internal key performance indicators. With an ERP system we buy it and implement it internally—it is within your four walls in terms of processes and functions.

The shared services model also provides the flexibility to match changing demand, another component of agility.

If we built the infrastructure in IBM, we cannot have the flexibility we need. We need to build the infrastructure size so it can meet peak capacity (for example quarter end order volumes) and if we do not have demand that utilises this capacity continually, we are not exploiting our infrastructure, but are still paying for it.

5.4. Measuring agility

Utilising an organisation whose core competence is in connecting up supply networks, rather than doing it themselves, enables IBM to significantly reduce the costs of connecting its systems to those of its customers and suppliers, and the time taken, once again contributing to increased agility.

The cost per connection is 50% lower for connecting and managing a new business partner via E2Open when compared to doing these activities ourselves.

As a result of adopting E2Open's services, IBM has reduced the time taken to respond to a customer order from over one hundred hours, to less than ten. The main reasons for this were the removal of people from the process and the elimination of batch processes, which previously ran only once a day. From an internal perspective, the elimination of manual 'touches' in the order-to-pay process is one of the main benefits that IBM attributes to adopting E2Open's services. For example, email orders that previously had to be manually entered into an ERP system are now passed directly from the systems of suppliers. This addresses a collective interest, "driving down the cost of these low value transactions, increases the profits for all of us and make us more competitive".

6. Discussion of case study findings: Themes for future research

Whilst case study research does not seek to be representative, the case undertaken suggests a number of interesting themes that deserve further investigation.

6.1. Redefining the relationship between IOS integration and flexibility

The first of these is that the open standards and hence the lower relationship-specific investments associated with the use of newer information system technologies such as web services should allow the trade-off that to date has existed between the degree of integration and the degree of flexibility that can be achieved in inter-organisational systems (Allen & Boynton, 1991; De Leeuw & Volberda, 1996; McLaren, et al., 2002) to be broken, as shown in Fig. 4. This suggests that organisations can form close relationships with their trading partners which when required by market changes, can be discontinued and other, equally close relationships formed. Such close but flexible relationships are inherent in the supply chain agility paradigm.

6.2. Implications for coordination mechanisms: A third way

An interesting consequence of such a change in relationship between integration and flexibility is on the impact this may have on the coordination mechanisms between firms. Transaction cost economics (Klein, Crawford, & Alchain, 1978; Williamson, 1975) describes two basic coordination mechanisms, markets and hierarchies. In their ‘move to the market’ hypothesis, Malone, Yates, and Benjamin (1987), suggest that increased use of inter-organisational information systems will result in a greater use of markets relative to hierarchies. In contrast, in their ‘move to the middle’ hypothesis, Clemons et al. (1993) assert that due to the existence of relationship specific investments, this move to the market will not occur, rather “the firm will rely on fewer suppliers than before, with whom the firm will have close and long-term relationships and with whom the firm will cooperate and coordinate closely”.

A redefinition of the relationship between integration and flexibility suggests that the logic of the “move to the middle” hypothesis, although well borne out to date, will no longer apply and

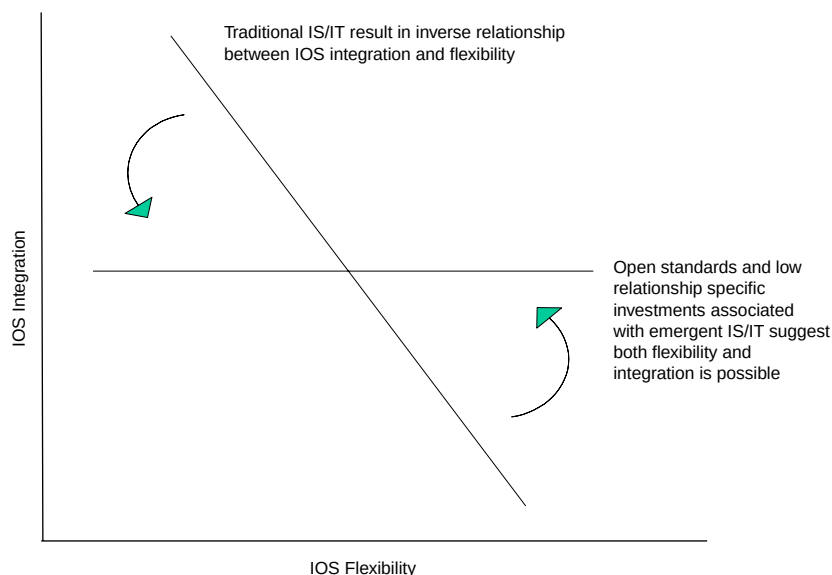


Fig. 4. Relationship between IOS integration and IOS flexibility.

the logic of Malone et al. should prevail. However, a move to increased use of markets is in contrast to the popular notions of improved supplier relationships and widespread supplier rationalisation initiatives (Christopher, 2000). A balance between the market approach and that of more enduring relationships, might be found in a third way, in which firms form close partnerships with suppliers and customers and then when market conditions dictate break those partnerships and form equally deeply integrated partnerships with other firms. Indeed, such behaviours are already being witnessed in a number of industries (Hertz, 2001). For example, White and Daniel (2004) describe how, in the automotive industry, the car manufacturers are seeking two types of suppliers. One group will be long-term partners, with whom they always expect to trade. The second group is partners who are involved in a specific development or programme. Whilst this programme is ongoing, they will work closely, and expect the partner to be integrated into the supply chain or network, however: “...when the OEM stops the programme, we will have to reconfigure the network, and look for the appropriate suppliers for the next challenge”.

6.3. *Increased outsourcing*

The hypotheses of both Clemons et al. and Malone et al., both suggest that increased use of information systems enables increased outsourcing of activities by firms. The case study findings are in agreement with this by demonstrating how IBM is prepared to outsource the key activity of developing and operating electronic trading links with their customers and suppliers to E2Open, rather than develop such solutions themselves. The willingness to outsource this activity was described as resulting from it not being viewed as a core competency of IBM,

E2Open has a core competency in doing the translations of data between organisations and it was not necessarily a core competency for IBM. Whilst we could have developed competency in this area, we decided that it was not where our resources should be applied. It was more important for our resources to be applied at the business process level and at the customer and supplier management level.

Also savings made by outsourcing this activity have been used to invest in the strategic aspects of the management of the supply chain, once again allowing a concentration on activities of most value to IBM,

The savings we have realised at the level of the infrastructure, we have now reapplied at the business level. The key here is to be able to switch the focus from operational to transformational activities.

6.4. *Renewed interest in the ASP model*

The case study findings also suggest that there may be renewed interest and opportunities for the ASP (application service provider) model. The managers interviewed during the case study were happy to refer to their use of applications offered by E2Open as on an ASP basis, which not

only offered the potential for such applications to become standard across the industry, as discussed above, but also reducing the cost of use,

We are using the Supply Chain Process Manager module on an ASP basis and there are reduced costs in terms of the development and maintenance of the application.

Whilst a part of the reason for the use a third party as a source of web services may be similar to the rationale underlying ASP usage, there are a number of differences. One difference is that the emphasis of E2Open is on the integration between organisations. In contrast most of the offerings of ASPs were, and continue to be, aimed at improving the IT provision within single organisations. However, similar issues that dampened the adoption of ASPs may also act to inhibit the adoption of third party web service providers, such as the limited viability of many business models, a lack of confidence from users in this new form of outsourcing and an inability of the service providers to explain in business terms the benefits their services offer (Currie & Seltsikas, 2001; Ekanayaka, Currie, & Seltsikas, 2002). Interestingly it was found during the case study that the ability for shared systems to connect whole supply networks, made the ASP model more compelling,

To me the original ASP model did not make sense. However, here we are going up the value chain. We are talking about Business Process ASPs and about something that is common to lots of people in a shared community. It is not something that I need, but about something that we all need.

7. Summary

Many organisations are recognising that increased volatility and turbulence are becoming standard characteristics of their underlying markets. Developing an agile supply chain that allows such organisations to meet the variations in demand at an acceptable level of cost is now a major focus of many leading organisations (Christopher & Towill, 2002a; Fisher, 1997; Mason-Jones et al., 2000). Van Hoek, Harrison, & Christopher (2001) identify four components of supply chain agility: customer sensitivity, virtual integration, process integration and network integration, all of which rely on the ability to easily and quickly be able to form deeply integrated information systems linkages with trading partners.

To date there has been a trade-off between the level of integration required to allow the process, virtual and network integration Van Hoek et al. identify as pre-requisites of agility and the flexibility to be able to form new linkages as market conditions dictate. The case study undertaken suggests that new information systems and technologies, particularly web services and their use with other supporting technologies such as e-hubs, offer an opportunity to re-define the relationship between system integration and system flexibility and allow the promise of close, yet dynamic linkages to be realised, a pre-requisite for the achievement of supply chain agility.

One intention of this paper is to seek to encourage the identification of further instances of the use of emergent information systems and technologies to enable supply chain agility. By synthesising such examples, it will be possible to establish how such systems, either singly or in combination can provide the agility that organisations are seeking. This paper has also identified a

number of themes suggested by the case study and extant IOS literature that appear to warrant further study when undertaking such studies. These themes include a possible redefinition of the relationship between IOS integration and flexibility, implications for the coordination of trade and collaboration between organisations, a further increase in the outsourcing of activities and associated with this, a renewed interest in the ASP model, but this time with a focus on process ASPs for inter-organisational integration.

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Dr. Andrew White B.Sc., M.Sc., Eng.D is a Senior Research Fellow at Cranfield School of Management. His research interests are in the twin areas of the impact of information technology on business-to-business relationships, and the management of innovation. He is currently working on a number of research projects that examine the role of information systems in creating an agile supply chain. This includes technologies and services such as Electronic Marketplaces/Hubs, Web Services, RFID, and Business Process Management Tools. He is also conducting research into the management of innovation, particularly discontinuous innovation, and innovation at the level of inter-organisational relationships and networks.

Professor Elizabeth Daniel B.Sc., Ph.D., MBA is Professor of Information Management at the Open University Business School where she undertakes research and teaching in the fields of e-business and information systems. She has published a significant number of papers on these subjects in leading academic journals and a number of management reports. Prior to joining the OU, Elizabeth was a member of the Information Systems Research Centre at Cranfield School of Management. She has a first degree and Ph.D. in Physics and an MBA from London Business School.

Dr. Mohdzaher Mohdzain B.Sc., M.Sc., Ph.D. is a Research Fellow at Cranfield School of Management. Zaher's research focuses on information systems strategy in large organisations. He is currently conducting a research project on the role of emergent information systems in enabling supply chain agility, and another on programme management. Prior to joining Cranfield, Zaher first worked in an accounting firm, and later a commercial bank, before he started his academic career as a university lecturer. Zaher has a B.Sc. in Accounting and M.Sc. in Management Information Systems from Northern Illinois University, USA, and Ph.D. in Information Systems from Cranfield.